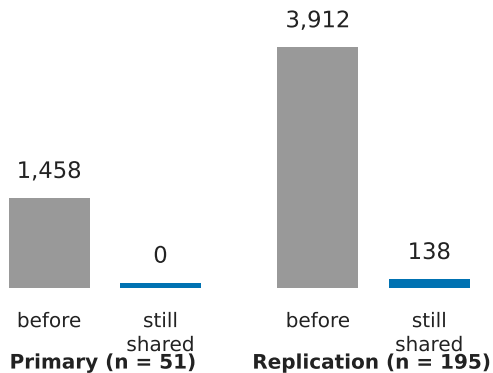


Is Alzheimer's disease-glioblastoma transcriptomic overlap real, or cell composition and age?

Bulk tissue (H1)

Shared AD-GBM DEGs, before and after adjustment for estimated cell composition



Cell-type level (H2)

Vulnerable
RORB+ neurons
(AD)

≠

Neural-like
malignant states
(GBM)

No overlap detected
(3 pre-specified tests; 3 donors per group)

Apparent convergence signal

Original analysis 6.54

+ donor modelling 7.25

+ age 1.29

+ donor + age 1.48

RRHO2 signal ($-\log_{10} P$); age and Braak stage
did not overlap in the discovery cohort

Shared bulk signatures track estimated cell composition; cross-disease comparisons need composition adjustment, donor-level inference and attention to age.